**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 304 · Final specification and synoptic checkpoint

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Name a PageRank property beyond counting inbound links.

Q2. What is a page fault?

Q3. What does an INNER JOIN return?

Q4. Search value 11 in [1,3,5,7,9,11,13], showing mid indexes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

synoptic — Combining knowledge across topics.

retrieval — Reproduce knowledge from memory.

misconception — Incorrect conceptual understanding.

intervention — Specific teaching targeting a gap.

### New learning

Use evidence rather than ticks alone — A topic is secure when a learner can solve an unfamiliar example, not merely recognise its name.

Find links between papers — Paper 1 numerical representation and database knowledge can underpin Paper 2 programming and algorithm questions.

Prioritise gaps — Record the exact misconception or missing step so later Paper 1/2 reteaching sessions target observable weaknesses.

### Worked example

Evaluate the claim 'all sorts take O(n log n) in the worst case'.

1. Predict first — Bubble sort has quadratic worst-case growth due to repeated adjacent comparisons.

2. Check reasoning — Merge sort has O(n log n) worst-case growth and generally needs O(n) auxiliary memory.

# We Do · Practise with guidance

Identify a paper-crossing weakness: a candidate knows binary search but cannot implement it.

```
Knowledge: sorted input and halving.
Implementation: low, high, mid updates.
Intervention: trace then code.
```

Predict first

Identify exactly which implementation state is wrong, e.g. high never decreases.

Check reasoning

Set a small test with a target below the midpoint to check the high-bound update independently.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State roles of PC, MAR, MDR, CIR and ACC, plus one bus role. [6]

### Question 2

A sorted list [2,4,6,8,10,12,14] is searched for 12. Trace binary search, then explain worst-case Big O. [6]

### Question 3

A school chooses AI scoring for pupil support. Give three contextual safeguards and explain risks. [6]

## Exit Ticket · Answer before checking

Knowing a topic name alone proves that you can apply the associated algorithm.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Weighted contributions from links, damping and iterative calculation.

Q2. Required virtual memory page not resident in RAM.

Q3. Rows matching the join condition.

Q4. mid3 (7) then mid5 (11) found; index5.

### You Do pages · Independent questions

Q1. PC next instruction address [1]; MAR address [1]; MDR transferred value [1]; CIR current instruction [1]; ACC intermediate result [1]; address/data/control bus role [1].

Q2. mid3 value8 [1], low4 [1], mid5 value12 [1], return index5 [1], O(log n) [1] from halving [1].

Q3. Check data quality/incompleteness [2]; privacy and lawful use [2]; human oversight and appeal/false-positive impact [2].

### Exit Ticket · Model answer

False. An unseen trace or solution provides better evidence.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.